SHETH L.U.J & SIR MV COLLEGE

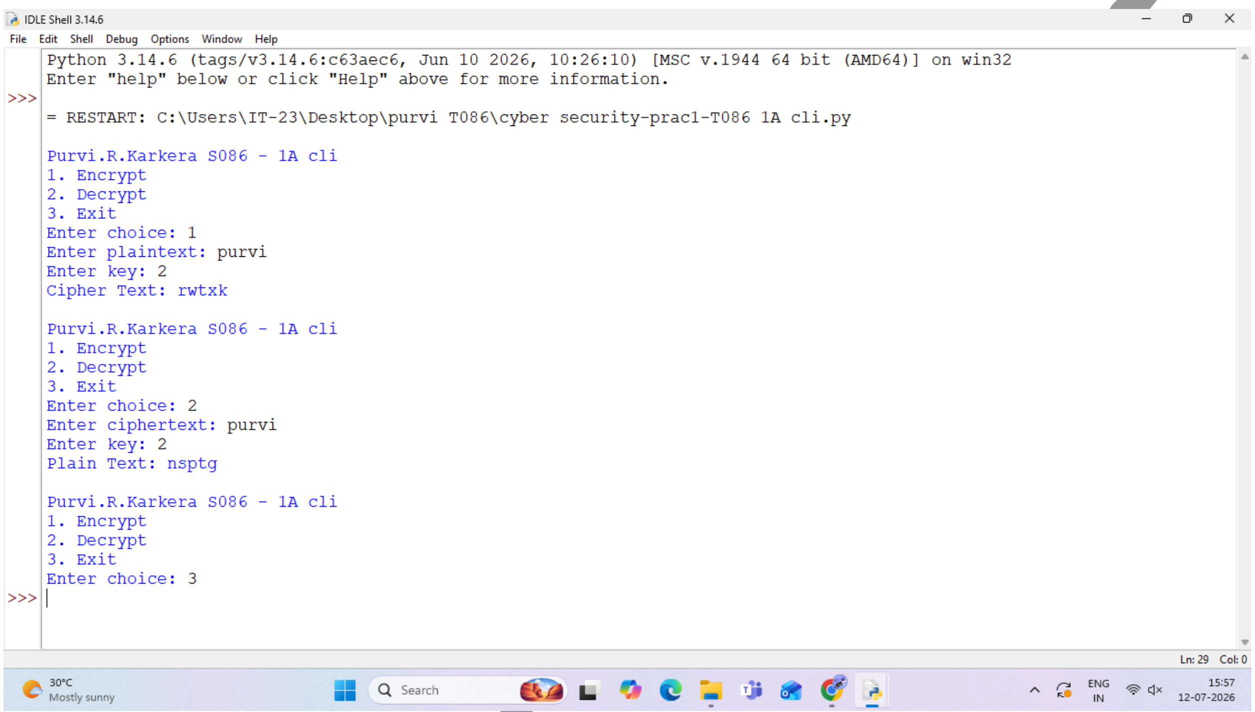
Practical 1: Implementing Substitution and Transposition Ciphers

1.A
Design and implement algorithms to encrypt and decrypt messages using classical
substitution techniques

CLI
CODE:
```
def encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - start + shift) % 26 + start)
        else:
            result += char
    return result
def decrypt(text, shift):
    return encrypt(text, -shift)
while True:
    print("\nPurvi.R.Karkera S086 - 1A cli")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")
    choice = input("Enter choice: ")
    if choice == '1':
        text = input("Enter plaintext: ")
        shift = int(input("Enter key: "))
        print("Cipher Text:", encrypt(text, shift))
    elif choice == '2':
        text = input("Enter ciphertext: ")
        shift = int(input("Enter key: "))
        print("Plain Text:", decrypt(text, shift))
```

```
    elif choice == '3':
        break
    else:
        print("Invalid Choice")
```

Output:



GUI

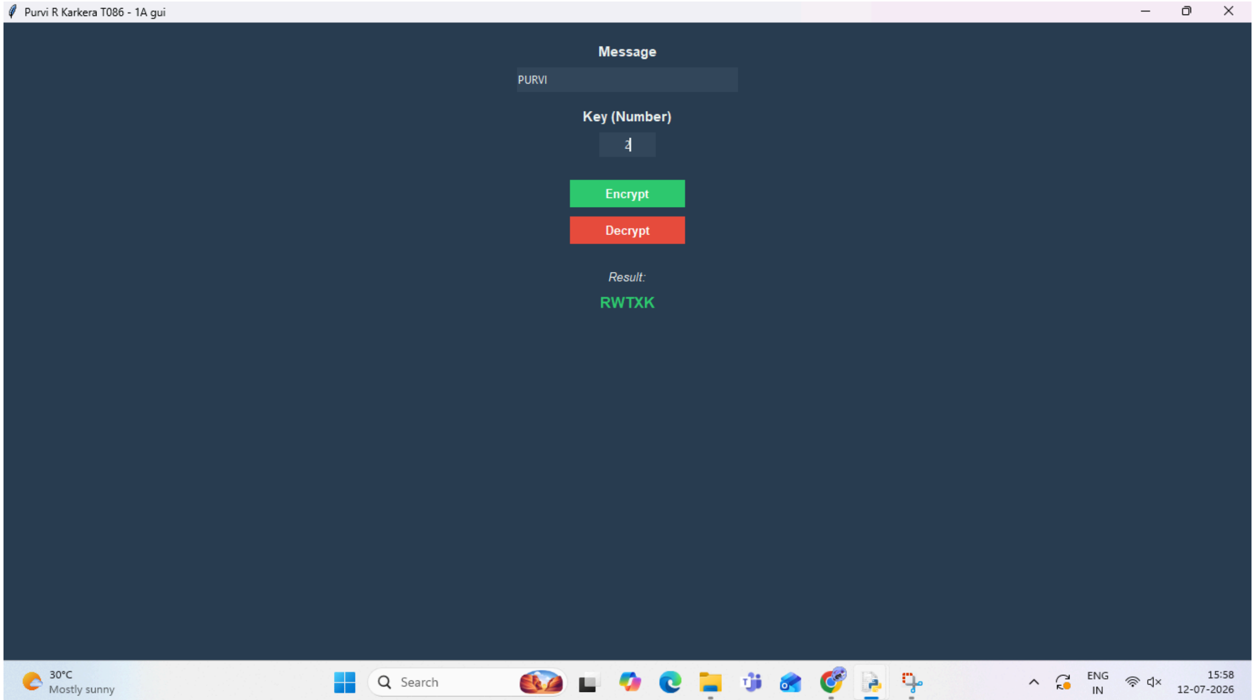
CODE:

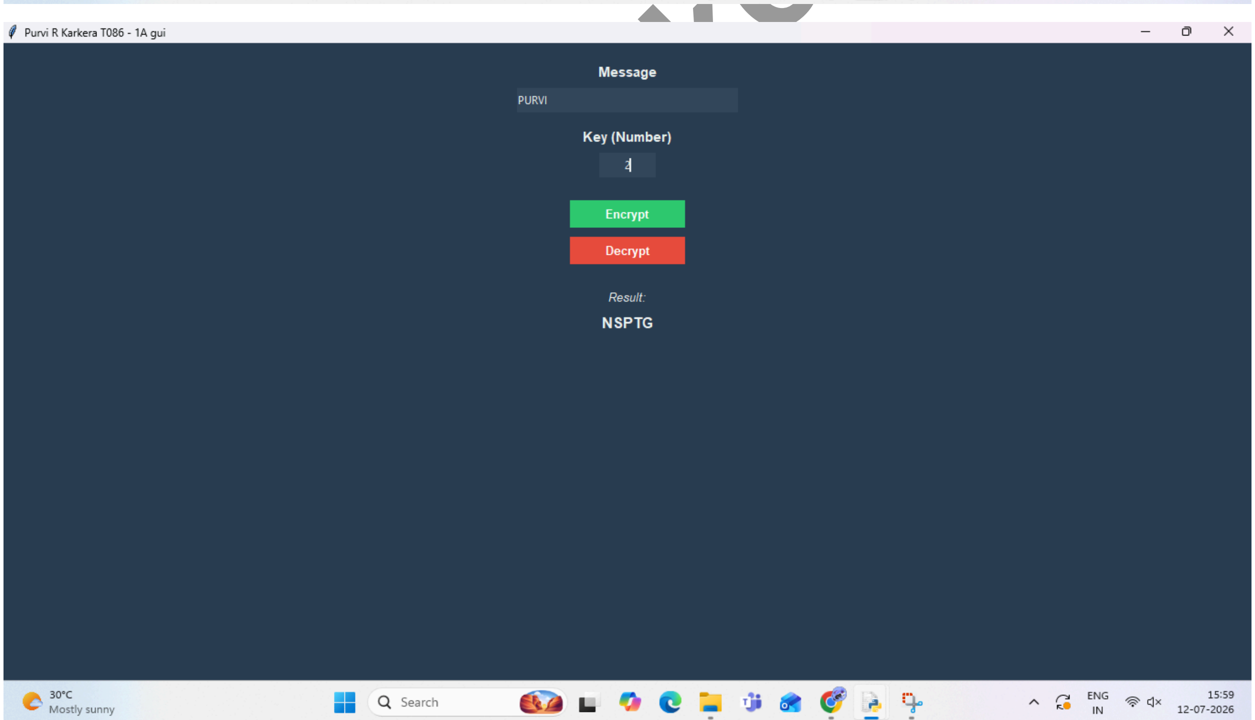
```
from tkinter import *
BG_COLOR = "#2C3E50"
TEXT_COLOR = "#ECF0F1"
ENTRY_BG = "#34495E"
ENC_BTN = "#2ECC71"
DEC_BTN = "#E74C3C"
BTN_TEXT = "#FFFFFF"
def encrypt():
    text = txt.get()
    try:
        shift = int(key.get())
```

```
    except ValueError:
        output.config(text="Please enter a valid number key!", fg=DEC_BTN)
        return

    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char)-start+shift)%26+start)
        else:
            result += char
    output.config(text=result, fg=ENC_BTN)
def decrypt():
    text = txt.get()
    try:
        shift = int(key.get())
    except ValueError:
        output.config(text="Please enter a valid number key!", fg=DEC_BTN)
        return

    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char)-start-shift)%26+start)
        else:
            result += char
    output.config(text=result, fg=TEXT_COLOR)
root = Tk()
root.title("Purvi R Karkera T086 - 1A gui")
root.config(bg=BG_COLOR, padx=20, pady=20)
Label(root, text="Message", bg=BG_COLOR, fg=TEXT_COLOR,
font=("Arial", 11, "bold")).pack(pady=(0, 5))
txt = Entry(root, width=40, bg=ENTRY_BG, fg=TEXT_COLOR,
insertbackground=TEXT_COLOR, bd=0, relief=FLAT)
```

```
txt.pack(ipady=5, pady=(0, 15))
Label(root, text="Key (Number)", bg=BG_COLOR, fg=TEXT_COLOR,
font=("Arial", 11, "bold")).pack(pady=(0, 5))
key = Entry(root, width=10, bg=ENTRY_BG, fg=TEXT_COLOR,
insertbackground=TEXT_COLOR, bd=0, relief=FLAT, justify="center")
key.pack(ipady=5, pady=(0, 20))
Button(root, text="Encrypt", command=encrypt, bg=ENC_BTN,
fg=BTN_TEXT, activebackground="#27AE60",
activeforeground=BTN_TEXT, bd=0, font=("Arial", 10, "bold"),
width=15).pack(pady=5, ipady=3)
Button(root, text="Decrypt", command=decrypt, bg=DEC_BTN,
fg=BTN_TEXT, activebackground="#C0392B",
activeforeground=BTN_TEXT, bd=0, font=("Arial", 10, "bold"),
width=15).pack(pady=5, ipady=3)
Label(root, text="Result:", bg=BG_COLOR, fg=TEXT_COLOR,
font=("Arial", 10, "italic")).pack(pady=(20, 5))
output = Label(root, text="", bg=BG_COLOR, fg=TEXT_COLOR,
font=("Arial", 12, "bold"), wraplength=300)
output.pack()
root.mainloop()
```

OUTPUT:


Purvi R Karkera T086 - 1A gui
Message
PURVI
Key (Number)
Encrypt
Decrypt
Result:
RWTXK
30°C
Mostly sunny
Search
ENG
IN
15:58
12-07-2026

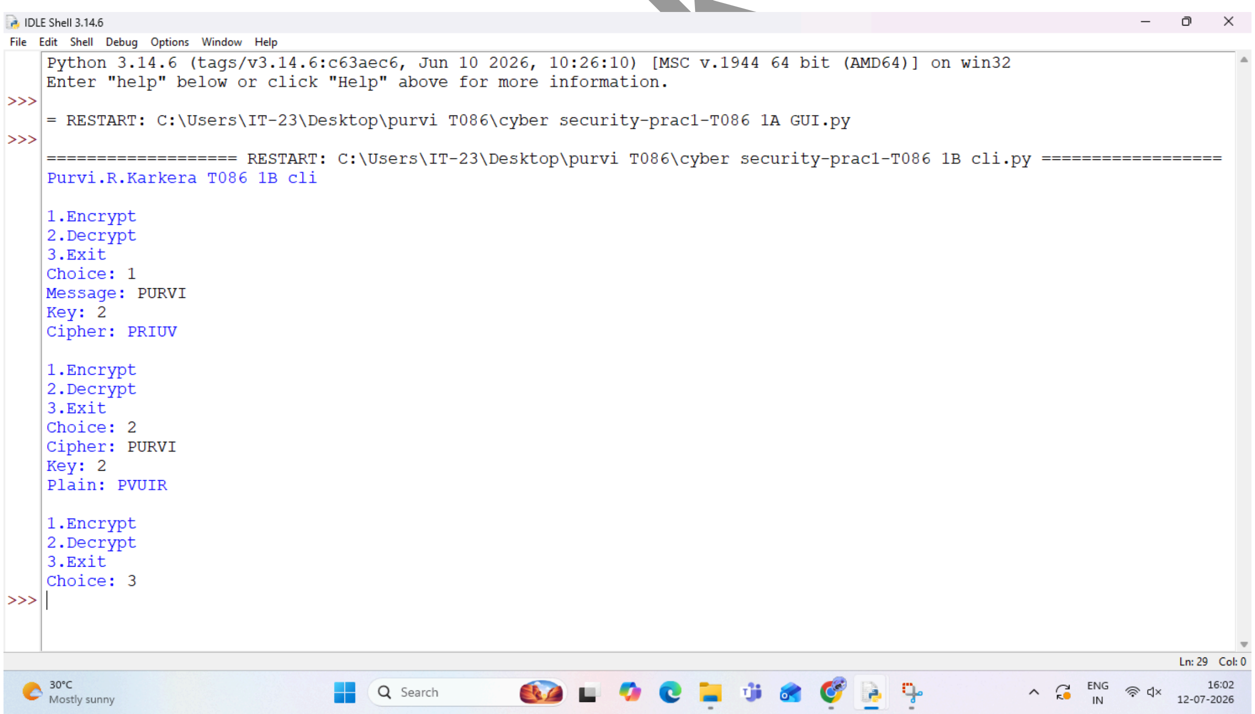

Purvi R Karkera T086 - 1A gui
Message
PURVI
Key (Number)
Encrypt
Decrypt
Result:
NSPTG
30°C
Mostly sunny
Search
ENG
IN
15:59
12-07-2026

1. B
Design and implement algorithms to encrypt and decrypt messages using transposition techniques

CLI
CODE:
```
import math
print("Purvi.R.Karkera T086 1B cli")
def encrypt(message, key):
    cipher = [''] * key
    for col in range(key):
        pointer = col
        while pointer < len(message):
            cipher[col] += message[pointer]
            pointer += key
    return ''.join(cipher)
def decrypt(cipher, key):
    num_cols = math.ceil(len(cipher) / key)
    num_rows = key
    shaded = (num_cols * num_rows) - len(cipher)
    plain = [''] * num_cols
    col = 0
    row = 0
    for symbol in cipher:
        plain[col] += symbol
        col += 1
        if (col == num_cols) or (col == num_cols - 1 and row >= num_rows - 
shaded):
            col = 0
            row += 1
    return ''.join(plain)
while True:
    print("\n1.Encrypt")
```

```
print("2.Decrypt")
print("3.Exit")
ch = input("Choice: ")
if ch == '1':
    msg = input("Message: ")
    key = int(input("Key: "))
    print("Cipher:", encrypt(msg, key))
elif ch == '2':
    msg = input("Cipher: ")
    key = int(input("Key: "))
    print("Plain:", decrypt(msg, key))
elif ch == '3':
    break
```


OUTPUT:




```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\IT-23\Desktop\purvi T086\cyber security-prac1-T086 1A GUI.py
>>>
================== RESTART: C:\Users\IT-23\Desktop\purvi T086\cyber security-prac1-T086 1B cli.py ==================
Purvi.R.Karkera T086 1B cli

1.Encrypt
2.Decrypt
3.Exit
Choice: 1
Message: PURVI
Key: 2
Cipher: PRIUV

1.Encrypt
2.Decrypt
3.Exit
Choice: 2
Cipher: PURVI
Key: 2
Plain: PVUIR

1.Encrypt
2.Decrypt
3.Exit
Choice: 3
>>> |
```

GUI

CODE:

```
from tkinter import *
import math
BG_DARK = "#1E1E24"
BG_CARD = "#2A2A32"
TEXT_LIGHT = "#E1E1E6"
TEXT_MUTED = "#A8A8B2"
ACCENT_CYAN = "#00ADB5"
ALERT_RED = "#FF2E63"
def encrypt():
    msg = txt.get()
    try:
        key = int(k.get())
        if key <= 0: raise ValueError
    except ValueError:
        output.config(text="Enter a positive number key!", fg=ALERT_RED)
        return
    cipher = [''] * key
    for col in range(key):
        pointer = col
        while pointer < len(msg):
            cipher[col] += msg[pointer]
            pointer += key
    output.config(text=''.join(cipher), fg=ACCENT_CYAN)
def decrypt():
    msg = txt.get()
    try:
        key = int(k.get())
        if key <= 0: raise ValueError
    except ValueError:
        output.config(text="Enter a positive number key!", fg=ALERT_RED)
        return
    num_cols = math.ceil(len(msg) / key)
    num_rows = key
```

```
    shaded_boxes = (num_cols * num_rows) - len(msg)

    plaintext = [''] * num_cols
    col = 0
    row = 0
    for char in msg:
        plaintext[col] += char
        col += 1
        if (col == num_cols) or (col == num_cols - 1 and row >= num_rows - shaded_boxes):
            col = 0
            row += 1
    output.config(text=''.join(plaintext), fg=TEXT_LIGHT)
root = Tk()
root.title("Purvi.R.Karkera T086 - 1B gui")
root.config(bg=BG_DARK, padx=25, pady=25)
Label(root, text="MESSAGE", bg=BG_DARK, fg=TEXT_MUTED, font=("Arial", 10, "bold")).pack(anchor="w", pady=(0, 5))
txt = Entry(root, width=40, bg=BG_CARD, fg=TEXT_LIGHT, insertbackground=TEXT_LIGHT, bd=0, relief=FLAT, font=("Arial", 11))
txt.pack(ipady=6, pady=(0, 15))
Label(root, text="KEY (NUMBER)", bg=BG_DARK, fg=TEXT_MUTED, font=("Arial", 10, "bold")).pack(anchor="w", pady=(0, 5))
k = Entry(root, width=15, bg=BG_CARD, fg=TEXT_LIGHT, insertbackground=TEXT_LIGHT, bd=0, relief=FLAT, font=("Arial", 11), justify="center")
k.pack(ipady=6, pady=(0, 20))
Button(root, text="Encrypt", command=encrypt, bg=ACCENT_CYAN, fg=BG_DARK, activebackground="#00888F", activeforeground=BG_DARK, bd=0, font=("Arial", 11, "bold"), width=15, cursor="hand2").pack(pady=5, ipady=4)
Button(root, text="Decrypt", command=decrypt, bg=BG_CARD, fg=TEXT_LIGHT, activebackground="#3E3E4A", activeforeground=TEXT_LIGHT, bd=0, font=("Arial", 11, "bold"), width=15, cursor="hand2").pack(pady=5, ipady=4)
```

```
Label(root, text="RESULT", bg=BG_DARK, fg=TEXT_MUTED,
font=("Arial", 9, "bold")).pack(pady=(20, 5))
output = Label(root, text="", bg=BG_DARK, fg=TEXT_LIGHT,
font=("Consolas", 13, "bold"), wraplength=300, justify="center")
output.pack()
root.mainloop()
```

OUTPUT:

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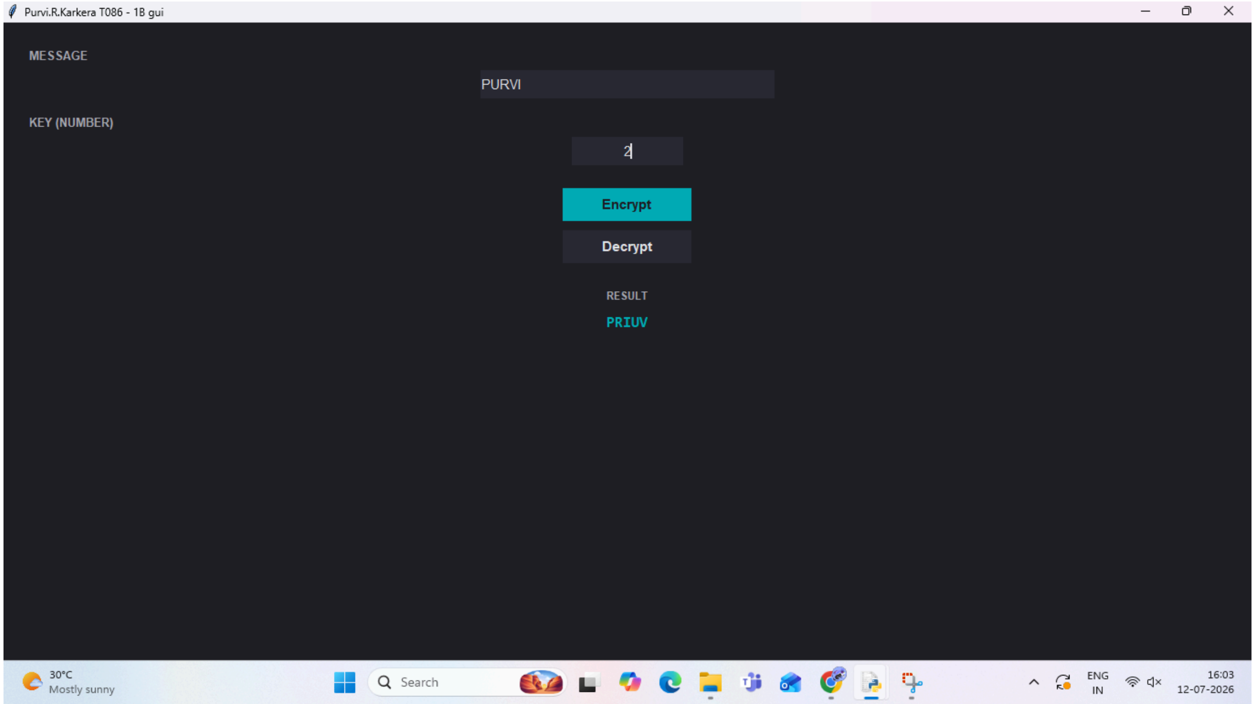

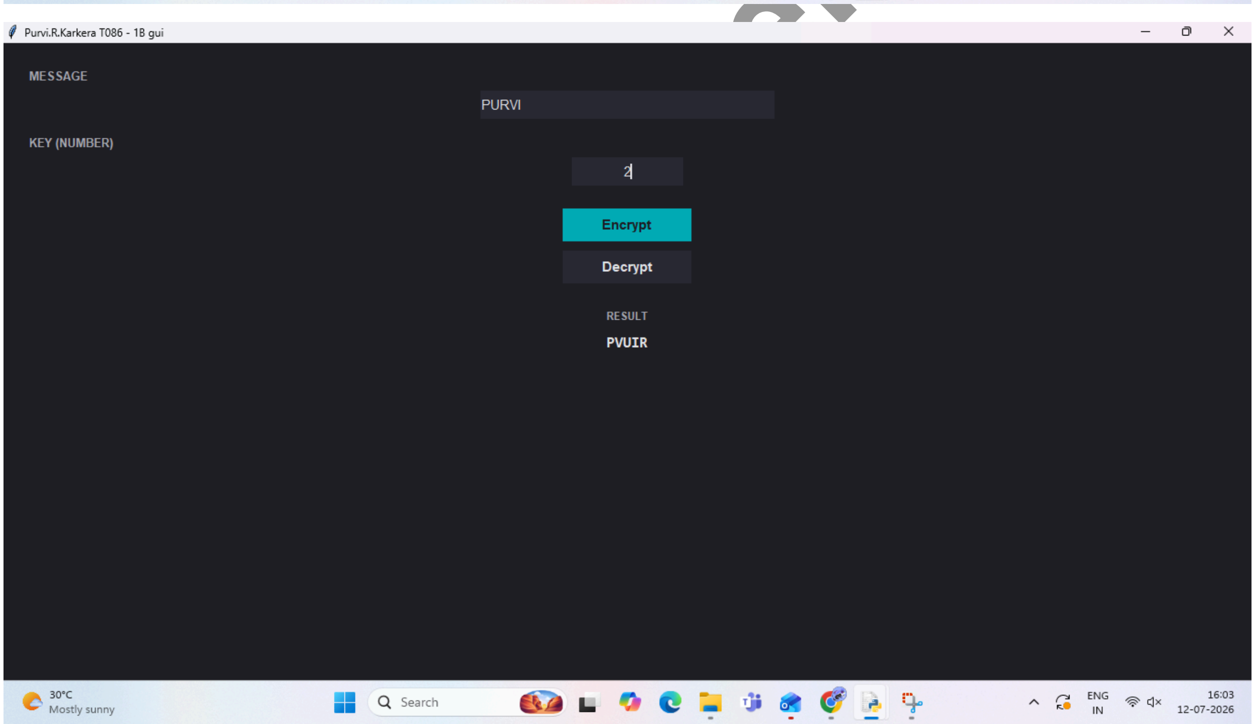

Purvi.R.Karkera
T086